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Abstract

LC-MS/MS-based proteomic analyses are widely used in biomarker studies because they can examine thousands of proteins simultaneously. However, missing values, batch effects, and data preprocessing decisions can seriously affect the analysis results in these data. This leads to the creation of false positive biomarker candidates and raises questions about the reliability of candidate proteins. In this project, a bioinformatics pipeline called RobuProt was developed to identify more reliable biomarker candidates in LC-MS/MS proteomic data by analyzing different scenarios and generating a robustness score.

RobuProt does not stick to a single analysis scenario. It evaluates the same dataset under different analytical conditions. In this context, it uses three different missing value completion methods: KNN, MinProb, and median imputation. Each method was run on datasets with and without batch correction, thus creating a total of six analysis scenarios. Each protein was scored using criteria such as significance frequency between scenarios, preservation after batch correction, missing value rate, fold change direction consistency, and ROC-AUC-based discriminative power.

The pipeline was tested using an open-access squamous cell carcinoma (SCC) proteomic dataset. After filtering, 3,103 proteins were used in the analysis. While the classical MinProb_noBatch method identified 260 significant proteins, RobuProt classified 13 of these candidates as highly reliable, 215 as moderately reliable, and 32 as possible artifacts. The results show that RobuProt distinguishes method-sensitive signals, moving more consistent and reliable biomarker candidates to higher positions in the final biomarker candidate list.

1. Introduction

1.1 Importance of Biomarker Discovery

Cancer is one of the leading causes of mortality and morbidity worldwide. Due to its heterogeneous nature, its progression through different metabolic pathways, and the variable responses it exhibits among individuals, the identification of biomarkers is of great importance in cancer research. Biomarkers are biological molecules that objectively measure normal or harmful biological processes or responses to treatment. They directly reflect various genes, proteins, metabolites, or cellular functions. Therefore, they play a significant role in translational research. Early diagnosis in cancer research is achieved through biomarkers. Detecting the disease in its early stages increases the success of treatment.

Another application of biomarkers is prognostication. This type of biomarker provides information about the potential for disease progression and the risk of metastasis. The behavior of some proteins, when overexpressed, becomes aggressive, while the decrease in others leads to poor clinical outcomes. Evaluating treatment response is also considered part of this process within biomarker research. The development of targeted therapies and immunotherapies, in particular, has increased the importance of molecular biomarkers. Based on these processes, the personalized medicine approach has come to the forefront. Personalized medicine determines the most appropriate and responsive treatment paths according to the patient's genetics. Identifying safe biomarkers is crucial for this approach. In line with this importance, biomarker research has been increasing rapidly. However, when looking at omics-based studies, the high-dimensional data obtained, technical reasons, missing values, and batch effects bring problems such as false positive biomarkers. As a result, biomarker discovery studies require not only experimental statistical analysis but also evaluation and research into analytical robustness and reproducibility. In recent years, LC-MS/MS-based technologies have begun to be used in biomarker discovery studies analyzing thousands of proteins. This also significantly changes the strategies used in proteomic data analysis. Therefore, the importance of reliable biomarker discovery is increasing.(Schwämmle, V, 2019; Shi, Y. 2025)

1.2 LC-MS/MS Proteomics and Computational Challenges

1.2.1 LC-MS/MS Proteomics

Proteomics is defined as the comprehensive study of all proteins found in a cell, tissue, or organism. Since proteins enable the execution of cellular functions, such analyses play a critical role in monitoring and understanding disease stages. For example, in cancer research, changes in protein expression play a role in biomarker discovery and the development of targeted therapy strategies.

In recent years, Liquid Chromatography–Tandem Mass Spectrometry (LC-MS/MS) technology has become the foundation of modern proteomics studies by providing high-capacity and high-sensitivity protein analyses. The LC-MS system, which has many examples in the literature and

has been increasing in recent years, consists of two basic stages. In the first stage, liquid chromatography (LC), proteins are enzymatically broken down into peptides. This mixture is separated by the liquid chromatography system. At this stage, peptides are separated from the clone at different times according to their charge, chemical properties, and hydrophobicity. Tandem mass spectrometry (MS/MS) is then used as the second stage. In the first stage, the mass/charge ratios of the peptides are measured. Then, the selected peptide ions are fragmented, and the fragment ions are analyzed. This analysis estimates the amino acid sequence in the peptide, and protein identification is performed. Thus, it is determined which protein it belongs to.

The most commonly used method in these studies is label-free quantification (LFQ). In this approach, the amount of protein is calculated by comparing the peptide densities measured among the samples. This method is preferred because it is flexible due to the need for an isotope or chemical labeling. However, it presents significant challenges in terms of analysis due to the high variation, missing values, and the need for extensive preprocessing. (Jin, L.,2021; Shi, Y. 2025)

1.2.2 High-Dimensional Data in Proteomics

Data obtained from LC-MS/MS analyses are presented as high-dimensional proteomics data. In the most common proteomic studies, thousands of proteins are measured simultaneously. However, the sample size remains quite limited. This situation is referred to in the literature as the "high-dimensional-low sample size" problem. It makes statistical analysis quite difficult.

This type of data also contains a high level of biological noise. Variables such as instrument sensitivity and ionization efficiency affect protein densities. Therefore, significant technical variations can occur even in samples with the same protein state. (Jin, L., 2021)

Another problem in the literature is data sparsity. In this data, many protein bases are not measured and are observed as missing values. This missing value problem is more common in proteins with low absorbance levels. In this way, the reliability of data matrices and downstream analyses decreases.

This data structure, which contains high noise and missing values, requires compitational preprocessing processes in proteomic analyses. Normalization, missing value padding, and data filtering are critically important methods. However, differences in these preprocessing methods can significantly alter the analysis results. (Yasui, Y., 2003)

1.2.3 Preprocessing in Omics Analysis

Preprocessing steps in omics data are one of the most important steps in biomarker discovery processes. Since raw data are not suitable for analysis, they need to undergo various preprocessing steps before analysis. These processes play a significant role in the analysis results. The first step is normalization. Normalization is a method aimed at reducing technical differences between samples.

Another step is missing value analysis. Missing values are quite common in this type of analysis. Therefore, methods such as NN, MinProb, QRILC, or median imputation are used to fill in these values appropriately. However, since these methods also have different assumptions, different analyses can be obtained from the same dataset.

Another important step is batch correlation. Batch effect refers to technical variations resulting from different analysis days, operation, and instrument calibration processes. When this situation is not corrected, technical differences lead to the emergence of false positive biomarkers in the literature.

Another important stage is preprocessing variability. Applying different preprocessing methods to the same dataset can profoundly change the analysis results. This leads to reproducibility problems in biomarker discovery studies.

Based on these reasons, biomarker selection based solely on statistical analysis is not considered sufficient in today's widespread literature. For a biomarker to be considered reliable, it must remain stable under different preprocessing conditions. Therefore, preprocessing-based studies are gaining importance. (Diao, X., 2025; Harris, L., 2023)

1.3 The Missing Value Problem in Proteomics

In label-free quantitative proteomics studies based on LC-MS/MS, missing values disrupt data integrity, creating methodological problems in terms of analysis reliability and interpretability. Webb-Robertson et al. reported that the rate of missing values at the peptide level in LC-MS proteomic datasets varies between 20% and 50%. This significantly exceeds the missing value rate, which is generally below 5% in microarray studies. Missing values not only affect data quality but also significantly impact post-imputation classification accuracy, variance measurements, and differential analysis results. (Webb-Robertson et al., 2015)

In their study, Lazar et al. stated that the missing value problem should not be evaluated solely based on numerical results; the mechanism by which the missing value occurs should also be examined. They demonstrated that multiple missing mechanisms can exist simultaneously in a dataset, and each mechanism may require a different imputation strategy approach. Therefore, it is accepted that analyses based on a single method, or a single imputation method, are not very reliable methodologically. (Lazar et al., 2016)

1.3.1 Sources of Missing Values

There are many different factors that cause missing values in LC-MS/MS proteomic experiments. These factors can come into play at different stages of the analysis, creating a complex deficiency in the data.

Webb-Robertson et al. have identified that missing values in label-free LC-MS experiments do not result from a single cause, but rather from multiple mechanisms. Low-abundance proteins are just one of these causes. Some proteins are undetectable because they fall below the

detection threshold. Because they are undetectable, low-abundance proteins are missed. This is called intensity-dependent censoring. Missing values in proteomic data are both more frequent and generated by more complex mechanisms compared to microarray-based gene expression data. (Webb-Robertson et al., 2015)

Jin et al. stated that protein deficiencies are generally evaluated in two categories: biological and analytical causes. Biological factors include the actual absence of the protein in the sample or the protein concentration being below the detection limit and the instrument being unable to measure it. Analytical factors include protein loss during sample preparation, mis-cleavage during digestion, poor ionization, and incorrect peptide-spectrum matching. Furthermore, because stochastic sampling methods are used in DDA (data-dependent acquisition) systems, different peptides are selected in each analysis. Therefore, different deficiency patterns can emerge even among technical repetitions. (Jin et al., 2021)

Wei et al. also stated that ionization-related deficiencies are observed in metabolomic data. Compounds below the detection limit are considered MNAR, random fluctuations in the instrument are considered MCAR, and deficiencies resulting from peak detection/co-elution errors are considered MAR. It is difficult to distinguish between MCAR and MAR deficiencies. This makes it difficult to select the appropriate imputation method in proteomic data. (Wei et al., 2018)

1.3.2 Missing Data Mechanisms

To properly analyze missing data, it is first necessary to understand the missing mechanisms that cause these missing values. In proteomic studies, missing values are examined in two categories: MAR and MNAR.

1.3.2.1 Random Missing (MAR)

Jin et al. define MAR, or random missing, as situations arising from technical limitations and stochastic fluctuations, independent of protein quantity. In such missing cases, although sufficient protein is present in the sample, it is not randomly detected in the analysis. The DDA method, by its nature, can cause MAR-type missing values. (Jin et al., 2021)

Wei et al., on the other hand, evaluated the MAR mechanism more broadly. According to them, the probability of a value being missing is determined by other observed variables in the dataset. For example, errors in the data preprocessing stage such as false peak detection and co-elution errors can be considered within the scope of MAR. They also stated that the distinction between MCAR and MAR is not clear in practice, and that some imputation methods can be applied to both MCAR and MAR type deficiencies. (Wei et al., 2018)

1.3.2.2 Non-Random Deficiency (MNAR)

MNAR is defined as a mechanism where missing values are directly related to the unmeasurable value itself. That is, some proteins cannot be measured because their quantities are much lower than the device's detection limit, even though the sample is present. Therefore, the deficiency does not occur randomly. The low amount of the protein is the cause of the missing value.

Webb-Robertson et al. stated that standard imputation methods designed for MAR cause significant performance loss when used for MNAR. Lazar et al. explained this situation with the concept of left censoring. According to this concept, proteins that are close to or below the device's detection limit produce a higher rate of missing values. These deficiencies are concentrated in the low-density part of the data distribution. This is a structural, not random, form of deficiency. (Webb-Robertson et al., 2015; Lazar et al., 2016)

This explains why standard imputation methods fail in proteomic data. Standard methods generally assume that deficiencies occur randomly and act accordingly. However, in the case of MNAR, deficiencies are caused by low amounts of proteins. Therefore, special imputation methods that consider the concept of left censoring should be used for MNAR-type deficiencies.

Jin et al. stated that MNAR is a significant source of protein deficiencies. They showed that imputation success depends not only on the overall deficiency rate but also on the MNAR rate. In this study, deficiency values generated with different MNAR rates were added to the dataset and compared with 7 different imputation methods. As the MNAR rate increased, the accuracy of all methods decreased. This decrease was not the same for every method. (Jin et al., 2021)

Wei et al. evaluated the QRILC method as suitable for MNAR-type deficiencies. This method models left-censored deficiencies. It is estimated that random sampling from a truncated distribution was used for MNAR deficiencies, while the random forest (rf) method showed better results for MAR or MCAR type deficiencies. (Wei et al., 2018)

1.3.3 The Impact of Missing Values on Differential Analysis

Improper handling of missing values can disrupt statistical analyses in proteomic data. Differential analysis, classification, and variance calculations are directly affected by missing values. How missing values are identified significantly impacts the accuracy and reliability of the analysis results.

In a study conducted on three different LC-MS datasets, Webb Robertson et al. reported that the chosen imputation method affected both variance measurements and classification accuracy. In this study, no imputation method yielded significantly better results than others across all evaluation criteria. Furthermore, it was observed that in some cases, direct classification without imputation yielded better results than classification after imputation. This result demonstrates that automatic imputation is not always beneficial in every dataset. (Webb-Robertson et al., 2015)

Jin et al. explained that the choice of imputation method significantly affects downstream analyses through the FADR metric. FADR is a criterion that shows the percentage of proteins

identified as misidentified proteins. In this study, among 7 different imputation methods, the random forest (rf) based imputation method was found to be the most successful. The RF method kept the FADR below 5% and detected true positives at the highest rate. Accordingly, simple left censoring methods (LOD, ND) could not adequately detect true positives. Methods that use the general structure of the dataset, such as SVD and BPCA, produced almost twice as much FADR as RF, thus not providing reliable results. (Jin et al., 2021)

Välikangas et al. evaluated five different software workflows on four different spike-in datasets. This study showed that the accuracy of differential expression detection with the amount of missing values varies from software to software. In terms of differential analysis, Progenesis software exhibited the most consistent performance. The combination of data filtering and local least squares imputation proved to be the best combination for improving overall analysis success. (Välikangas et al., 2016)

These findings suggest that missing values are not simply gaps that need to be filled. The choice of imputation method can directly and significantly affect differential analysis results, false positive rates, and the reliability of biological interpretations. Therefore, in proteomic studies, the imputation method should be analyzed in conjunction with data preprocessing steps.

1.3.4 Imputation Strategies in Proteomics

Different imputation methods are used to fill missing values in proteomic data. Webb-Robertson et al. divided these methods into three groups: single-value filling methods, local methods based on similar samples or proteins, and global structure-based approaches. This classification helps determine which mechanism causes missing values and which method should be preferred, as not every imputation method is equally successful for every type of deficiency. (Webb-Robertson et al., 2015)

1.3.4.1 K-Nearest Neighbor Imputation (KNN)

KNN imputation uses the most similar neighboring values in the dataset to predict the missing value. Webb-Robertson et al. applied this method based on Euclidean distance. They used the 10 nearest neighbors for each peptide as a local similarity method. If the missing value is present in all of these neighbors, the method moves to the next 10 neighbors. (Webb-Robertson et al., 2015)

KNN is particularly successful in MAR-type cases where the deficiency is random. KNN does not show sufficient success in left-censored MNAR-type deficiencies.

Jin et al. also included the KNN method in their comparative analysis. They noticed that the error rate in this method increased as the MNAR ratio increased. In particular, the prediction success of KNN was quite low when the MNAR ratio was 50% and 80%. Therefore, KNN should not be automatically preferred for every dataset. (Jin et al., 2021)

1.3.4.2 MinProb/QRILC Imputation

The QRILC method is a technique specifically developed for MNAR-type deficiencies caused by low abundance. Wei et al. reported its success. In the QRILC method, missing values are estimated from the low-density portion of the data distribution. For proteins considered to be below the device detection limit, missing values are not randomly completed. Instead, they are completed by modeling from the low-value region. (Wei et al., 2018)

Wei et al. compared eight imputation methods on different metabolomic datasets. The QRILC method produced the lowest error value under MNAR conditions. PCA and PLS-Procrustes analyses also confirmed that the QRILC method gave the best result and best preserved the inter-sample distribution. (Wei et al., 2018)

Lazar and Burger developed the MinProb method within the imputeLCMD package. In this method, missing values are completed by randomly selecting from a Gaussian distribution formed around the lowest values observed in the dataset. Thus, it is predicted that the missing values originate from a region with lower biological density. Lazar et al. explained that MinProb is more efficient than MAR methods, especially in MNAR-dominant datasets and protein-level analyses. (Lazar et al., 2015)

1.3.4.3 Median Imputation

Median imputation fills in missing values with the median values in the dataset. It belongs to the single-value substitution group in the classification by Webb-Robertson et al. It is a frequently used method because of its simplicity. However, its simplicity does not necessarily mean it is reliable or always usable.

Wei et al. compared median imputation with eight methods and stated that this method showed moderate, stable performance in MCAR and MAR type deficiencies. However, in MNAR-dominant left-censored conditions, median imputation was not as successful as methods like QRILC. Because while MNAR deficiencies are expected to mostly occur in the low-density region, median imputation fills these deficiencies with the mid-values of the dataset. This reduces the success rate. (Wei et al., 2018)

Furthermore, the variance in the dataset may decrease when using methods that fill with constant values like the median. This can lead to both missing true positives and false positives in differential analyses. Therefore, median imputation is generally not used as the primary solution method in current proteomic analyses. It can be considered a reference method used to compare the performance of newer methods.

1.3.5 Limitations of Single Imputation Approaches

The fact that missing values in proteomic data arise for various reasons indicates that analyses based on a single imputation method do not yield reliable results.

Webb-Robertson et al. concluded that no single imputation method offers a general solution due to complex deficiency mechanisms in LC-MS proteomic data. The same method was successful

in some criteria but unsuccessful in others. This suggests that using a single method is risky and misleading. (Webb-Robertson et al., 2015)

Lazar et al. also emphasized that multiple deficiency mechanisms can be present simultaneously in the same dataset. Instead of asking "which is the best general imputation method?", the question should be "which method is best according to the deficiency structure in this dataset?". For MAR, the MAR method should be used, and for MNAR, the MNAR method should be used. Furthermore, it was noted that whether the imputation is performed before or after the peptide harvesting step significantly affects the results. (Lazar et al., 2016)

Jin et al. stated that the methods highlighted in different studies are not always parallel to each other. This contributes to the reproducibility problem in the literature, because the list of biomarkers changes depending on the imputation method used. (Jin et al., 2021)

Välikangas et al. took a broader perspective. They showed that not only the imputation method but also other pipeline steps such as normalization, quantification, and data filtering affect the results. This finding indicates that an optimization encompassing the entire pipeline is necessary. (Välikangas et al., 2016; Välikangas et al., 2017)

For these reasons, relying on a single imputation method in proteomic analyses is very risky. Comparing different imputation scenarios and evaluating how consistent the results are between methods may be safer. The multi-imputation scenario-based robustness score pipeline designed and developed within the scope of this project is expected to provide a solution to this need.

1.4 Batch Effect in Omics Data

1.4.1 Batch Effect

Batch effects represent technical variations that occur independently of biological variables in omics studies. These problems arise due to different instrument settings, operator changes, sample preparation at different times, or changes made in the laboratory. Batch effects appear to be a significant problem, especially in high-data sets. In LC-MS/MS instrument analyses, when samples belonging to the same biological group are analyzed with different batch effects, samples can be separated based on technical similarities, and their true biological relationships can be distorted. This situation can lead to misinterpretations in protein analysis and biomarker studies. (Goh, W., 2017)

Another important feature is that it is present in some samples in a hidden way. Although it appears as a biological difference in the analysis, it is often caused by a technical problem. This is critically important for correction using appropriate methods. (Yu, Y., 2023)

1.4.2 Sources of Batch Variation

This arises from various variables at different stages of the proteomic analysis process. These problems frequently occur in LC-MS/MS studies and directly affect data quality and results. (Zhang, Y., 2018)

1.4.2.1 Acquisition Date

Analyzing samples on different days is one of the most common causes of batch variation. LS devices may undergo small changes over time. These small but significant changes in ionization efficiency, column age, and device sensitivity directly affect signal intensities. This results in different intensity profiles for the same biological samples on different days. Therefore, the reliability of the results decreases. And it directly affects the signal results. (Bessa, C., 2017)

1.4.2.2 Operator Differences

Operator differences are based on differences in sample preparation and equipment use in different studies on the same dataset. Although laboratory changes such as sample processing may seem like minor differences, they can significantly alter protein concentrations. This is especially noticeable in high-sensitivity proteomic analyses. Therefore, the reliability of the results is reduced in this way. (Van Ooijen, M., 2018)

1.4.2.3 Chromatography Drift

Performance changes in the column involved in liquid chromatography are defined as chromatography drift. Changes in column pressure, solvent and solute composition, or retention time cause peptides to appear at different concentrations in analyses. Therefore, the reliability of the results changes. (Skoraczyński, G., 2022)

1.4.2.4 Sample Preparation Variability

The protein preparation stages, such as protein extraction, cleaning, digestion, and peptide preparation, are considered the most sensitive stages of the analysis. Minor changes made during sample preparation can significantly alter protein concentration and output. (Zheng, R., 2021)

1.4.2.5 Instrument Calibration and Sensitivity Changes

Mass spectrometers are among the instruments that require continuous calibration. Since small changes in instrument sensitivity directly alter the results, regular monitoring is necessary. Furthermore, factors such as detector performance and ionization stability also contribute to varying protein concentrations in the same samples. (Cheng, W. L., 2023)

1.4.3 Impact of Batch Effect

Batch effect leads to significant analytical problems in studies. When technical problems are not corrected, biological signals can be misinterpreted. This can also lead to the emergence of incorrect biomarker candidates. (Zhang, Y., 2018)

1.4.3.1 False Clustering in PCA

One of the most common results is the separation of samples according to technical batches rather than biological groups. In unsupervised analyses such as PCA or hierarchical clustering, while separation of samples according to patient/control/status is expected, separation according to the day of analysis or the device is observed. This result significantly reduces the reliability of the biological output. (Che, Y., 2024)

1.4.3.2 Artificial Group Separation on Biomarker Discovery

Batch effect can sometimes create false group separations. This situation, which is quite common when control and patient samples are analyzed in different batches, also reduces the reliability of biological data. (Liu, Q., 2020)

1.4.3.3 False Differential Proteins

When batch correction, which is applied to and aimed at improving the batch effect, is not applied, some proteins may be considered significant only due to technical differences. This situation, especially during biomarker discovery, leads to the emergence of incorrect biomarker candidates and subsequently to incorrect results.It also seriously affects clinical implications.

For these reasons, the batch effect is not only a technical problem but also a critical analytical problem affecting the accuracy and reliability of biological interpretations. (Yu, Y., 2023)

1.4.4 Batch Correction Methods

Due to the many negative effects of the batch effect, batch correction methods have been developed in the literature. These methods aim to reduce technical problems and preserve biological signals. (Yu, Y., 2023)

1.4.4.1 Mean-Centering Approach

This approach uses the mean-centering method. The average of the protein concentrations within each batch is calculated, and systematic shifts are corrected according to the batch.

This method is simple and applicable. It provides effective results in small datasets, and at the same time, it has achieved success in reducing technical problems by comparing before and after results. (Čuklina, J., 2021)

1.4.4.2 ComBat and Empirical Bayes Approaches

One of the most commonly used methods to reduce the batch effect is 1.4.4.2 ComBat and Empirical Bayes Approaches. This approach involves systematic modeling of batch-related problems. Empirical Bayes-based methods show effective performance, especially in large omics datasets. The ComBat method is frequently used in the field of omics in the literature and significantly reduces the impact of the batch effect on the downstream data. However, excessive application of these methods can lead to the loss of real biological signals. Therefore, these

strategies should be applied carefully, and the preservation of biological variation should be the primary goal. (Zhang, Y., 2018)

1.5 Reproducibility Challenges in Computational Proteomics

1.5.1 Analytical Variability Between Analysis Workflows

In LC-MS-based proteomic analyses, results are not only determined by the raw data obtained. The pipeline used during the analysis can significantly affect the results. The method used to complete missing values, the chosen normalization method, whether batch-effect correction is applied, and the type of statistical test used can change the resulting protein list. One of the biggest problems here is that different imputation methods predict the cause of missing values differently. For example, some methods say that the deficiency occurred randomly, while others say that the deficiency occurred because low amounts of proteins were not detected by the detector. Therefore, the chosen imputation method affects which proteins are found to be statistically significant. (Lazar et al., 2016; Webb-Robertson et al., 2015)

Similarly, the normalization method also changes the analysis results. Välikangas et al. demonstrated that a single normalization method does not yield the best results in all different datasets. This prevents accurate comparison of results from different studies. (Välikangas et al., 2016)

Consequently, the same raw dataset yields different biomarker lists when analyzed using different pipelines. This difference may not always stem from true biological differentiation. In some cases, differences in methodological approaches in the analysis also significantly affect the results. (Jin et al., 2021)

1.5.2 False Positive Biomarker Discovery

Using only a single pipeline in proteomic studies leads to the occurrence of false positive biomarkers. False positive biomarkers are proteins that appear as significant in the analysis results even though they are not actually related to the disease. This occurs when missing values are completed using methods incompatible with the data structure. MNAR-type missing values are known to originate from low-abundance proteins. If a pipeline is used that completes this type of deficiency with methods that assume random deficiencies, it distorts the folding rate result of some low-abundance proteins. This leads to erroneous differential expression signals. (Lazar et al., 2016; Webb-Robertson et al., 2015)

Beyond data completion methods, problems arising from sample preparation and sample quality also affect biomarker analysis. For example, erythrocyte contamination, changes in platelet count, or differences in coagulation status can give the impression that there are genuine biological differences between the case and control groups.

Geyer et al. draw attention to the need to be mindful of such sample-related deviations in biomarker studies, as these differences can be mistakenly considered as disease-related signals. (Geyer et al., 2019)

Therefore, using false positive biomarker candidates in validation studies can lead to both time and financial losses. Furthermore, this situation is among the significant factors that make it difficult to transfer proteomic biomarkers to clinical practice.

1.5.3 Need for Robustness Assessment

Proteomic analysis results can be sensitive to the pipeline used. Therefore, it is not sufficient for a protein to be found significant as a biomarker based on the results of only one analytical method. To obtain more reliable results, the results of different analytical conditions should also be examined to see if they yield consistent results.

For example, if a protein is significant only in a single imputation and normalization combination but not when analyzed using other appropriate methods, this result is considered analytically weak. Such a result may be due to the chosen analytical method rather than a true biological signal. (Jin et al., 2021)

Therefore, multi-scenario evaluation is crucial in proteomic studies. Whether proteins are significant regardless of the method when different imputation, normalization, and statistical analysis options are tried should be examined. Sticker et al. have shown that more robust peptic and protein-level summarization methods can reduce false discovery rates and improve folding change predictions. (Sticker et al., 2020)

Therefore, robustness assessment based on multiple scenarios is not an additional or superfluous analysis in proteomic biomarker studies. It is an important method for leading to more reproducible, reliable, and clinically applicable results.

1.6 Limitations of Single-Pipeline Biomarker Discovery

1.6.1 Classical Biomarker Discovery Workflow

Classical biomarker discovery studies described in the literature conduct analysis processes through a single data processing pipeline. Thus, raw LC-MS/MS data is first subjected to quality control and filtering, and then the resulting missing values are filled in using a single imputation method. After this process, differential potential protein analysis is performed between patient and control groups. Proteins that meet a certain threshold value become biomarker candidates.

This approach aims to be practical and fast. And this type of approach is successful. Especially in big data processing, researchers choose a single imputation method to facilitate the analysis process. For example, after the missing values are filled in using methods such as KNN or MinProb, they are sorted according to p-value or FDR values.

Although this approach is successful, it has some limitations and shortcomings. The resulting list of biomarker candidates is entirely dependent on a single analytical strategy. That is, even if a protein is significant in one analysis, it may not be significant in another. This discrepancy is often not tested, therefore the classical pipeline may be questionable in terms of biological reliability and may be inadequate. (Čuklina, J., 2021; Hernández, B., 2014)

1.6.2 Dependence on Analytical Decisions

Proteomic analyses depend on many decisions. Determining the normalization method when filling in missing values, correcting the batch effect, and testing its applicability directly affect the results. Therefore, the same LS set can yield different results with different analytical decisions.

In particular, the imputation method directly affects the ranking of biomarkers. When the KNN method is examined, it mainly uses the similarity between proteins. MinProb, on the other hand, counts those with missing values from a lower density. In this way, the two methods produce different biomarker candidates.

When looking at batch correction, it has a very high effect. When the batch effect is not corrected, technical problems appear biological, leading to false positive biomarkers. When batch correlation is applied, some of these proteins often lose their significance. This explains why the classical and single-method pipeline approach is susceptible to technical problems.

When all these processes are examined, the concept of "method-sensitive proteins" stands out. This concept refers to proteins that appear significant only under a specific pretreatment or analytical condition, but do not remain stable under different analytical methods.

Therefore, some classical biomarkers are not true biomarkers but rather biomarkers that are highlighted and nominated as candidates based on analytical decisions. Biomarkers with a large missing value and batch effects should be re-examined before being accepted as reliable biomarker candidates**.** **(**Jiang, B., 2023; Gregori, J., 2012; Ivanisevic, J., 2019)

1.6.3 Lack of Reliability Scoring

In biomarker discovery studies in the literature, proteins are generally evaluated using statistical measures such as p-value or FDR. However, these measures do not directly show how stable a protein remains under different pretreatment conditions. In other words, a protein being statistically significant does not directly indicate its analytical stability.

Most studies in the literature also fail to calculate biomarker candidates as a separate robustness score or robustness metric. Therefore, the confidence level of a protein is generally not clearly classified. For example, if a protein has a low p-value, is significant only with a single imputation method, and has high batch correlation or missing value rate, this protein may have lower reliability in practice. Classical pipelines are limited in their ability to bridge this gap.

The lack of reliability scoring is a significant problem in translational or clinical biomarker studies because carrying false positive candidates to further validation studies directly leads to cost and time loss. However, reporting these types of biomarkers is extremely important in terms of reproducibility and prediction in different studies. Therefore, in biomarker discovery studies, simply distinguishing between meaningful and meaningless results is not enough to contribute to the literature. A robustness score is needed that evaluates the stability of candidate proteins under different analytical conditions and their ability to be preserved after batch correlation, along with the missing value ratio. Such approaches are not a replacement for classical statistical approaches, but rather an addition to them as a prerequisite for quality and reliability. (Adriani, M., 2020)

1.7 Need for Robustness-Based Biomarker Selection

1.7.1 Concept of Analytical Robustness

In biomarker studies, a protein being found to be statistically significant alone is insufficient to demonstrate that it is a reliable biomarker. This is because LC-MS/MS data also include differences such as missing values, batch effect imputation, and normalization methods. Therefore, for a biomarker to be considered a reliable candidate, it is expected to remain stable in different data processing methods. This concept is defined in the literature as analytical robustness.

Analytical robustness refers to the capacity of a result to be preserved under different computational methods. From a proteomic perspective, it is expected that a protein will be preserved similarly in different imputation methods and batch correlation strategies. If a protein is significant only in one imputation method and loses its significance in other methods, this protein may be method-sensitive, rather than a biologically strong candidate. Reproducibility indicates the capacity to be reproduced. It is expected that a biomarker candidate identified in a study will yield similar results under different analytical conditions. However, due to the large number of studies in the literature using classical and single-pipeline methods, the focus is often on a single analytical result. This prevents directly measuring the level of reproducibility.

Analytical stability, on the other hand, examines whether a protein remains stable within the same analysis. In the literature, if a protein shows similar results before and after batch correlation in different imputation methods, it is seen to have higher analytical stability. Therefore, it is used as an important criterion in the confidence ranking in biomarker candidate selection. (Ferreira, S. L., 2017)

1.7.2 Multi-Scenario Validation Strategy

In the Robustness approach, biomarker selection means evaluating the same dataset by processing it through multiple preprocessing steps instead of relying on a single analytical result. This approach is defined as the Multi-Scenario Validation Strategy. The primary goal of

this strategy is to test how stable biomarker candidates remain under different conditions and to find the best candidate.

This study is based on the Multi-Scenario Validation Strategy. The same LC-MS datasets are analyzed in parallel using different imputation methods and batch correlation conditions to find the best candidate. Thus, instead of relying on a single preprocessing decision, each preprocessing condition is examined.

The multiple imputation strategy is one of the foundations of this study. The reasons for missing values in proteomic data are not always the same. Sometimes they are due to technical malfunctions, and sometimes to malfunctions in the biological data. Therefore, a single imputation method reduces reliability and is replaced by the multiple imputation strategy. The parallel use of different imputation methods such as KNN, MinProb/QRILC, and median imputation allows testing for differences in missing values.

Comparing batch corrected and uncorrected analyses is also a step in this process.Analyses without batch correction represent the classic pipeline output. Batch-corrected analyses, on the other hand, measure which proteins remain significant after technical problems are reduced. If a protein appears significant only in the uncorrected analysis and disappears after batch correction, it may be a candidate for a false positive due to the batch effect.

Therefore, multi-scenario validation offers a more reliable filtering approach for biomarkers. It shows not only the significance of each protein but also the degree of significance under different conditions. It also allows for viewing and comparing results after correlation and with different imputations. (Ishikawa, R., 2021; Wryk, G., 2024)

1.7.3 Robustness Scoring as a Reliability Layer

Robustness scoring is a reliability layer added to the classical and applied statistical biomarker selection. This layer combines the concepts of significance and robustness. This concept should actually be both statistically significant and consistently prioritize in different analysis scenarios. Therefore, robustness scoring looks not only at significance but also at analytical reliability.

In the study conducted, a protein robustness score between 0-10 is calculated. The score looks at how many analyses the protein was significant in and its retention after batch correlation. Thus, a comprehensive reliability assessment is aimed for each protein.

Confidence-based biomarker prioritization, on the other hand, shows not only significance but also robustness scores, classifying a protein into classes such as reliable, moderately reliable, or highly reliable. The main advantage of this approach is that it does not delete or replace classical statistical data, but adds to it. This allows researchers to control the results more reliably. This provides a significant advantage in terms of candidate selection for clinical biomarker identification and further validation studies. (Jin, L., 2021; Brendel, W., 2019)

1.8 Aim of Study

Liquid chromatography-tandem mass spectrometry (LC-MS/MS) is a powerful analytical technique frequently used in proteomic studies. It allows for the simultaneous examination of numerous proteins thanks to its label-free quantification approach. Therefore, LC-MS/MS-based proteomic analyses are a very suitable method for identifying biomarker candidates, especially in cancer research. However, obtaining reliable biomarkers from this data does not depend solely on the measurement technique. The choices made during the data analysis process also significantly affect its reliability.

The aim of this study is to develop a robust analytical pipeline to increase the reliability of biomarker discovery in LC-MS/MS-based proteomic data. In classical proteomic analyses, biomarker candidates are determined using a single data preview and statistical analysis method. However, according to information obtained from the literature, analytical decisions such as how missing values are completed, the normalization method, batch effect correction, and the statistical test used determine which proteins will be found to be significant in the analysis results.

The primary goal of this project is to examine how consistent proteins found to be significant remain when different analytical methods are tested. Therefore, the designed pipeline does not rely on a single analytical pathway but compares different imputation and batch correction scenarios. This allows for the identification of protein candidates that remain consistent under different conditions, not just a specific method.

Another objective of the study is to reduce false positive biomarker candidates caused by missing imputation and batch effects. Defining a protein as a biomarker candidate based on a single significant analytical result is neither reliable nor accurate. Therefore, a robustness score was added to our pipeline for each protein, calculating its consistency under different analytical scenarios.

In general, this project does not aim to determine biomarker candidates in LC-MS/MS proteomic analyses solely based on statistical significance. It also evaluates them based on their consistency under different methodological conditions. The designed pipeline contributes to making existing differential analysis methods more reliable, reproducible, and leading to the discovery of more biologically significant biomarker candidates.

2. Market Gap Identification

2.1 Current Biomarker Discovery

In recent years, proteomic methods involving LC-MS/MS have become frequently used in biomarker discovery. In academic research laboratories, hospitals, pharmaceutical development companies, and research centers, diseases have begun to be examined in more detail thanks to the simultaneous measurement of thousands of proteins without waiting in line. The most common biomarker method in recent years consists mostly of data preprocessing, missing value filling, differential analysis, and selection of statistically significant proteins. These workflows are mostly performed with software-based analysis packages such as R/Python in the literature. The resulting protein lists are compared with statistical measures such as p-value and FDR. This approach, which has been used for a long time, is still not evaluated in terms of its reliability or stability under different analytical conditions.

Although the current method aims to find more significant proteins, there is no system that measures and analyzes the robustness of the found proteins against technical decisions. (Hernández, B., 2014)

2.2 Limitations of Proteomics Pipelines in Current

Although existing biomarker discovery tools are powerful, they have several limitations. Many analytical pipelines undergo only one preprocessing step chosen by the user. The extent to which results change when the imputation method, batch correlation method, and statistical filters used are modified is not examined.

Another significant problem is that the analysis results are directly dependent on the software used. Therefore, different biomarker name lists can be generated on the same dataset using different analytical methods. This leads to reproducibility problems.

Furthermore, most of the pipelines used present results as significant or insignificant. However, another deficiency identified here is the inability to determine the extent to which proteins found to be significant are significant and how much they are affected by changes. As a result, proteins that appear significant in the analysis are actually only a result of the chosen analytical method. (Hernández, B., 2014)

2.3 Missing Reliability in Current Approaches

The most commonly used statistical analysis methods in the literature are p-value, fold change, and FDR measures. These measures examine other groups of a protein and track whether there is a difference. Although this method is useful and important, it does not show the reliability of the protein and its sensitivity to other methods. Most analysis methods cannot answer the high rate of missing values, the change in the result when batch correlation is applied, or the fact that a protein remains significant in different imputation methods. These unanswered questions play

an important role in biomarker discovery studies in the literature. Accordingly, there is currently no mechanism that provides a significant-reliable assessment in proteomic analyses. This creates a significant gap in the literature.

2.4 Need in Biomarker Discovery

One of the biggest gaps in the literature on proteomic biomarker discovery is the lack of objective evaluation of the reliability levels of biomarkers identified as candidates, regardless of the method used. This is because researchers want to examine not only whether a biomarker candidate is significant, but also its stability in different methods. Reducing false positives, especially for candidate biomarkers to be used in the clinic, is very important for human health. Eliminating low-reliability candidates from the outset provides significant advantages in terms of time and cost.

Furthermore, this need affects not only the literature and researchers, but also biotechnology and pharmaceutical companies that serve humanity on a large scale. Selecting reliable biomarkers can also increase the success of translational research. (Adriani, M., 2020)

2.5 Market Opportunity for Biomarker Selection

In recent years, the proteomic data analysis market has grown rapidly worldwide. At the same time, the need for bioinformatics solutions is also increasing due to the use of LC-MS/MS. Existing software focuses on large datasets and analysis processes, limiting the scope of biomarker reliability. A bioinformatics platform that identifies biomarker candidates with established robustness scores and comparisons across various analyses could become a valuable tool not only for academia but also for research hospitals, biotechnology companies, and subsequently, pharmaceutical companies. The proposed this analysis aims to support existing workflows, not replace them. This system aims to provide a layer of reliability in addition to existing pipelines, thus ensuring more secure decisions and outcomes.

3. Proposed Solution: RobuProt

3.1 Product Overview

RobuProt is a bioinformatics pipeline designed to provide a robustness score to improve the reliability of biomarker candidates in LC-MS/MS-based proteomic data. Considering the fundamental problems explained in the Introduction section, supported by literature, the need for a solution is evident, particularly regarding the impact of missing values on differential analysis results, the potential for batch effects to create erroneous biological interpretations, and the fact that using a single pipeline produces method-based results.

Therefore, RobuProt is not simply a classic pipeline that finds meaningful proteins. It is a pipeline that evaluates the consistency of proteins using different analytical scenarios, providing more reliable biomarker candidates.

3.2 Scientific Concept

The fundamental concept of RobuProt is to evaluate the same dataset using different scenarios without relying on a single analytical output. This approach relates to the statement emphasized in section 1.3.5 that relying on a single imputation method is risky. Furthermore, as mentioned in section 1.5.1, analyzing the same raw dataset with different pipelines is highly likely to generate different biomarker lists.

Based on this, it compares the results obtained under different imputation methods and batch correction conditions. Thus, it can evaluate whether a protein is a truly reliable candidate or merely a protein that appears significant depending on the method in a particular analysis result.

3.3 Innovative Aspect and Originality

RobuProt's unique aspect is that it analyzes different scenarios simultaneously and adds a robustness score to the resulting biomarker selection. As mentioned in section 1.6.3, classical biomarker discovery studies generally evaluate using statistical measures such as p-values or FDRs. These measures do not show how stable a protein remains under different data preprocessing conditions.

RobuProt fills this gap by not evaluating each protein solely based on statistical significance. It evaluates it based on its consistency in different scenarios. Thus, biomarker discovery moves away from a risky selection process based on a single significance result. It transforms into a reliability-based biomarker discovery.

3.4 Competitive Advantages

RobuProt's most significant advantage is its ability to detect and eliminate false positive biomarker candidates at an early stage. As explained in section 1.5.2, inappropriate data completion methods or sample-related deviations can cause proteins that are not actually related to the disease to appear as biomarkers.

RobuProt tests candidate proteins with multiple scenarios to mitigate this risk. This ensures careful and safe selection of proteins to be carried over to further biological or clinical processes. This creates a significant advantage in terms of both time and cost.

3.5 Workflow Architecture

The RobuProt workflow begins with the acquisition, filtering, and application of different analysis scenarios of the LC-MS/MS proteomic dataset. The dataset is completed with different imputation methods. Each output is analyzed with and without batch correction. In the next stage, differential expression analysis is performed for each scenario. The resulting lists of significant proteins are compared. A stability score is calculated for each protein. In the final RobuProt, proteins are ranked as biomarker candidates with high, medium, or low reliability according to their scores.

3.6 Expected Benefits Compared to Classical Pipelines

The main advantage of RobuProt compared to classical pipelines is that it highlights not only meaningful proteins but also analytically stable proteins. This results in a more reliable, interpretable, and clinically suitable list of biomarkers.

In general, RobuProt aims to reduce false positives caused by missing values and batch effects in LC-MS/MS proteomic analyses, increase reproducibility, and perform biomarker discovery using a more reliable method.

4. Scientific Principles Behind RobuProt

4.1 The Basis of Missing Value Imputation

RobuProt's missing value approach is based on the effects of the missing value problem, described in section 1.3, on biomarker discovery. Specifically, the distinction between MAR and MNAR in 1.3.2 clarifies that missing values are not a uniform problem. Therefore, RobuProt does not use a single imputation method as a solution.

With this principle, it applies different imputation methods in separate analysis scenarios. The aim is to check whether the significance of a protein is associated only with a specific missing value completion method. This reduces the differential analysis biases caused by imputation, as described in sections 1.3.3 and 1.3.5.

4.2 The Basis of Batch Effect Correction

RobuProt's batch effect approach stems from the idea, described in section 1.4, that technical variations can mislead biological interpretations. As stated in 1.4.3, the batch effect can cause samples to cluster according to technical groups, artificial group separation, and the formation of false differential proteins.

Therefore, RobuProt performs analyses with and without batch correction. If a protein appears significant only before correction and not after correction, this candidate should be considered a false positive due to batch correction. Proteins that remain consistent after correction are considered more reliable and are prioritized in the candidate list.

4.3 Differential Expression Analysis Basis

RobuProt uses differential expression analysis to identify significant protein differences between biological groups. However, as a final mechanism, RobuProt does not use this analysis alone.

This is because, as described in 1.5.1 and 1.6.2, data preprocessing decisions can alter the final protein list. Therefore, RobuProt does not generate differential analysis results with only one scenario. It compares and evaluates them under different imputation and batch correction conditions.

4.4 Robustness Assessment Basis

In RobuProt, robustness assessment forms the core scientific decision-making logic. According to the concept of analytical robustness described in 1.7.1, a reliable biomarker candidate is expected to remain stable in every computational analysis.

This principle is applied to each protein in RobuProt. If a protein is significant in many scenarios, it is considered a more robust and reliable biological signal. Conversely, if it is significant in only one scenario, the method is considered a susceptible candidate.

4.5 Confidence-Based Biomarker Prioritization

In the final stage, RobuProt classifies candidate proteins according to their reliability levels. This approach is related to the robustness scoring logic described in section 1.7.3. Accordingly, candidate proteins are not evaluated solely on whether they are significant or not. They are classified as high, medium, or low reliability candidates based on their consistency in different analysis scenarios.

This classification more clearly shows which proteins are more suitable for further biological interpretation or clinical studies. Thus, RobuProt transforms the classic statistical significance result into a more interpretable and reliable biomarker rating system.

5. Product Development Plan

5.1 Feasibility Assessment

5.1.1 Literature Review

LC-MS/MS proteomic biomarker discovery, missing value, and batch effect methods have been extensively reviewed in the literature. As a result of the review, it was found that most of the existing research relies on a single preprocessing method and, consequently, does not include a system that evaluates the reliability of biomarker candidates, and also demonstrates a reliability-aware biomarker selection approach in line with the needs.

5.1.2 Bioinformatic Software Evaluation

Various R/Python-based softwares were evaluated. It was determined that while the existing software used has strong differential analysis and big data processing capabilities, a robustness score that measures the robustness of biomarkers under different preprocessing conditions and an analysis method based on this score have not been produced.

5.1.3 Technical Studies

The applicability of all methods used has been demonstrated with the Python open-source library. This situation suggests that this implemented and developed analysis process is carried out at a very low cost.

5.2 Prototype Development

5.2.1 Pipeline Design

The planned pipeline includes data preprocessing, missing value analysis, batch effect evaluation, differential analysis, and robustness scoring steps.

5.3 Future Development Opportunities

5.3.1 Computational Design

It is planned to increase processing performance for processing larger datasets.

5.3.2 User Interface Simplification

The command lines and version are intended to be user-friendly. The aim is to minimize human error.

5.3.3 Scalability

It is planned to run and improve proteomic data formats and multi-omics datasets.

5.4 Applications of Bioinformatics Development

It can be used to support biomarker selection in academic fields, clinical research, and the biotechnology and pharmaceutical sectors.

6. Material Method

6.1 Data Set Selection

In this study, an open-access LC-MS/MS proteomic dataset comparing squamous cell carcinoma (SCC) with a control group was used. The dataset was obtained from the StatOmics/pda repository on Github

(https://raw.githubusercontent.com/statOmics/pda/data/quantification/cancer/proteinGroups.txt). The data used was quantified using a label-free method.

This dataset was chosen because it is suitable for cancer biomarker research. Furthermore, the presence of missing values in the dataset and the ability to separate samples into different batch groups made it suitable for testing and evaluating the developed pipeline.

6.2 Data Collection

The proteinGroup.txt file used is the standard output file of the MaxQuant software. This file was read in Python and the analysis was parsed as a tab-separated table. Columns starting with "Intensity" were extracted to obtain protein intensity values. A total of 18 samples were obtained in this way.

Group information was determined according to the sample names. Samples whose names started with "OR" were set as the disease group, and the rest as the control group. Batch information was also determined according to the sample order, i.e., the first 6 samples were Batch 1, 7-12 were Batch 2, and 13-18 were Batch 3. Then, a metadata table containing the sample name, group information, and batch information was created.

6.3 Data Filtering

Protein IDs were first retrieved from the "Majority protein IDs" column. If this information was not available, the "Protein IDs" column was checked. Intensity values that appeared as zero in the MaxQuant output were converted to NaN values to separate those that were truly zero from the noise.

A $\log_2$ transformation was applied to all intensity values. This transformation makes the data more suitable for analysis. If inf or undefined values occur in the data, it converts them back to NaN.

Additionally, proteins with a "+" in the "Potential contaminant" column were considered contaminants and excluded from the analysis. Finally, the missing value rate was calculated for each protein. If more than 60% of the samples contained missing values, those proteins were excluded from the analysis. This prevents proteins with a large amount of missing data from significantly affecting the results.

6.4 Missing Value Analysis

To visualize the distribution of missing values in the dataset, missing value rates were calculated both on a protein basis and a sample basis. The protein-based rate shows how many samples a protein was not measured in. The sample-based rate shows how many proteins were missing in a sample.

Missing values were visualized using a bar graph. Proteins were grouped according to their missing values. A heatmap was created by selecting the 60 proteins with the highest missing values. In this heatmap, samples were sorted by batch and group. This made it possible to see whether or not there was a relationship between missing values and a specific batch or group.

6.5 Batch Effect Evaluation

The batch effect was evaluated using PCA analysis. PCA analysis was performed both before and after batch correction. KNN imputation was applied for missing values before PCA. In the graphs, samples are color-coded according to disease and control groups. Batch information is indicated with different symbols.

This analysis allowed us to evaluate whether the samples were separated according to biological group or due to technical effects (batch effect). The PCA graph after batch correction was also used to see how the correction affected the samples.

6.6 Data Normalization

Median centering was performed on the filtered $\log_2$ intensity matrix. For this, the median of the values of the non-missing proteins in each sample was calculated. Then, it was subtracted from all values in that sample. This was done to reduce the overall intensity differences between samples. The matrix resulting from normalization was used as the base dataset in subsequent analyses.

6.7 Parallel Imputation Strategy

In this study, a single imputation method was not used to complete missing values. KNN, MinProb, and median imputation methods were used. Each method was applied to both the unbatched and batch-corrected datasets. This resulted in a total of six analysis scenarios: KNN_noBatch, KNN_Batch, MinProb_noBatch, MinProb_Batch, Median_noBatch, and Median_Batch.

These six scenarios were then compared in differential expression analysis and robustness score calculation. This evaluated whether a protein maintained its significance under different analysis conditions.

6.7.1 KNN Scenario

KNN imputation was applied using the "KNNImputer" function from the "scikit-learn" library. The number of neighbors was set to k=5. Missing values were estimated by considering the similarity structure between samples. KNN imputation was applied to both unadjusted and adjusted datasets. KNN_noBatch and KNN_Batch scenarios were created.

6.7.2 MinProb Scenario

MinProb imputation was applied to proteins that might originate from low-abundance proteins. For each protein, the 1st percentile μ of the observed values was used, and σ was 0.3 times the standard deviation. The minimum σ value was determined as 0.01. Missing values were filled with values randomly drawn from the $N(\mu, \sigma^2)$ distribution. For proteins with fewer than 2 observed values, the missing values were equated to the global minimum of the dataset.

This method was also applied to unadjusted and adjusted datasets to create MinProb_noBatch and MinProb_Batch scenarios.

6.7.3 Median Scenario

In median imputation, missing values were filled with the median values of each protein in the non-missing samples. This method was performed for comparison purposes. It was applied to both uncorrected and corrected datasets to create Median_noBatch and Median_Batch scenarios.

6.8 Batch-Corrected and Uncorrected Analyses

Each imputation method was applied to two different datasets: a normal dataset without batch correction and a dataset with batch correction. Uncorrected analyses represent the classic pipeline approach. Batch-corrected analyses were used to examine whether proteins retained their significance after the technical batch effect was reduced.

A mean centering approach was used for batch correction. The within-batch mean of proteins was calculated for each batch. The overall protein mean calculated from all samples was added back. This aimed to reduce batch-related systematic differences.

6.9 Differential Expression Analysis

The differences in protein abundance between disease and control groups were examined separately for each of the six analysis scenarios. The $\log_2$FC value for each protein was found by subtracting the control group mean from the disease group mean.

Welch's two-sample t-test was used for statistical comparison. A protein was considered significant in a given scenario if it met both $p < 0.05$ and $|\log_2 FC| > 1.0$. The MinProb_noBatch scenario was used as a reference for the classic pipeline comparison.

6.10 Multiple Test Correction

After calculating the p-values for each protein in each scenario, Benjamini-Hochberg FDR correction was applied. FDR values are shown in the results tables.

However, in this pipeline, the significance label is not directly based on FDR. It is determined according to the criteria of $p < 0.05$ and $|\log_2 FC| > 1.0$. FDR values were calculated for more transparent reporting of results. They are also included for additional statistical evaluation.

6.11 Robustness score Calculation

In this project, a robustness score between 0 and 10 was calculated for each protein. The aim of this score is to evaluate how consistently and reliably a protein provides results across different analysis outcomes.

Before scoring, contaminants and reverse databases starting with CON__ and REV__ were excluded from the analysis. Five criteria were defined for the robustness score, and their sum gives the robustness score.

Criteria 1 - Frequency of Significance Among Scenarios (0-3 points)

This is calculated based on the number of scenarios in which a protein is found to be significant. If it is significant in at least four out of six scenarios, it receives 3 points; if it is significant in two or three scenarios, it receives 1 point; and if it is significant in fewer than two scenarios, it receives 0 points.

Criteria 2 - Preservation After Batch Correction (0-2 points)

If a protein is significant in at least two out of three batch-corrected scenarios, it receives 2 points; if it is significant in only one batch-corrected scenario, it receives 1 point. If it is not significant in any batch-corrected scenario, it receives 0 points.

Criteria 3 - Missing Value Rate (0-2 points)

The missing value rate per protein has been calculated. A missing value rate below 30% receives 2 points, between 30% and 50% receives 1 point, and above 50% receives 0 points.

Criteria 4 - $\log_2$FC Direction Consistency (0-2 points)

The $\log_2$FC direction in each scenario has been compared for each protein. For this criterion to be calculated, the protein must have a valid $\log_2$FC value in at least four scenarios. Proteins with the same $\log_2$FC direction in all scenarios receive 2 points, proteins showing a different

direction in only one scenario receive 1 point. If there are more directional changes, the protein receives 0 points.

Criteria 5 - Discrimination Power with ROC-AUC (0-1 point)

The ability of each protein to distinguish between disease and control groups has been calculated using ROC-AUC. A data matrix obtained with KNN imputation was used for this calculation. Proteins meeting the condition of max(AUC,1 - AUC) > 0.75 receive 1 point.

6.12 Biomarker Classification Criteria

Proteins are examined in three groups according to their total robustness score. Those with a score of 7 and above are classified as High confidence, those between 4-6 as Medium confidence, and those below 4 as Possible artifact.

This classification was not done solely to distinguish between significant or insignificant candidate proteins. It was also done to evaluate them according to their reliability levels. Proteins in the high confidence group were consistent in different analysis results. Candidate proteins in the medium confidence group maintained their significance under some conditions. Those in the possible artifact group appeared to be more method-dependent or of low reliability. Therefore, they were removed from the final biomarker list.

The classic pipeline result and the robustness score pipeline result were compared. The MinProb_noBatch scenario was selected as the classic pipeline. Thus, the number of proteins included in the biomarker candidate list in classical analysis that remained at high and medium confidence was evaluated.

6.13 Statistical Software and Packages

All analyses were performed in Python3. Data processing was performed using pandas and NumPy libraries. KNNImputer from the scikit-learn library was used for KNN imputation. PCA analysis was performed using PCA and StandardScaler, both included in scikit-learn.

For differential expression analysis, the Welch t-test in scipy.stats was used; for multiple FDR correction, the multipletests and BH method in statsmodels were used; and for ROC-AUC calculation, the roc_auc_score in sklearn.metrics was used.

Graphs were created with matplotlib. As a result of the analysis, PCA graphs, missing value distribution graphs, heatmaps, volcano plots, robustness score graphs, and a comparison graph of the robustness score with classical pipeline were generated. The result tables were saved in CSV format. To ensure reproducibility, the random seed value was fixed at 42.

7. Results

7.1 Dataset Characteristics

This study used open-access SSC (Squamous Cell Carcinoma) data. The dataset contains a total of 18 samples. Nine of these are disease groups and the other nine are control groups. A total of 389 protein groups were identified. After quality control and filtering, proteins unsuitable for analysis were removed. This reduced the number of proteins to 3179. Subsequently, an incomplete data filtering strategy was applied, further reducing the number of proteins to 3103. The equal number of disease and control groups allows for better analysis of both groups.

Table 7.1 Dataset Characteristics

Characteristic	Value
Total number of samples	18
Disease group (SCC)	9
Control group	9
Initial number of protein groups	3,898
Proteins remaining after filtering	3,179
Proteins included in the final analysis	3,103

7.2 Data Quality Assessment

7.2.1 Missing Value Analysis

The missing value problem, which we identified as a problem at the beginning of the project, was examined before the analysis.

This missing value analysis revealed a heterogeneous distribution among the proteins.

1,897 proteins contain less than 10% missing values. 605 proteins contain more than 30% missing values. The distribution of missing values is shown in Figure 7.1.

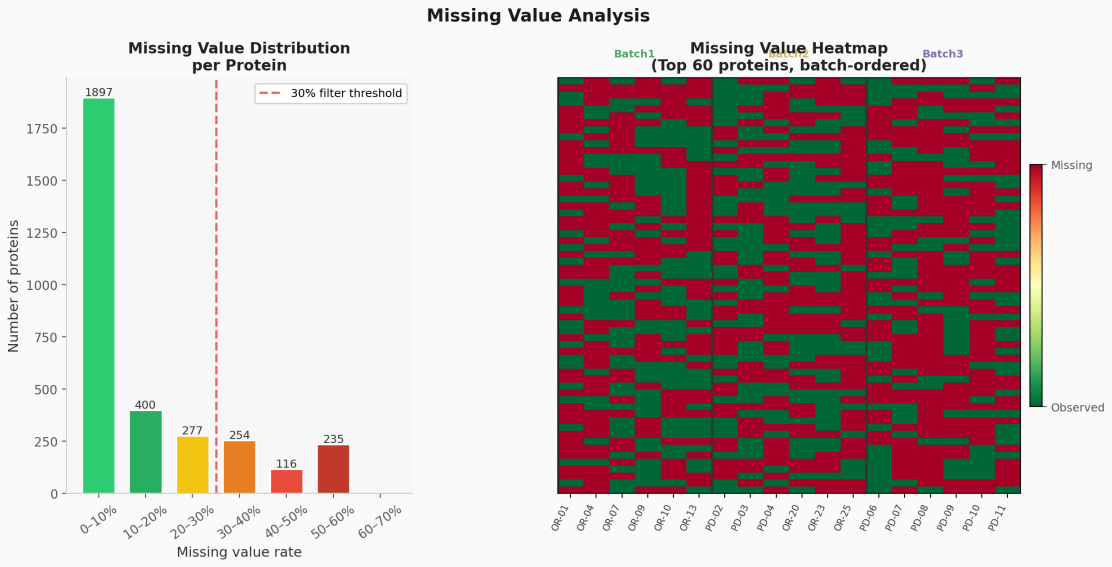


Figure 7.2.1. Missing value distribution per protein (left) and heatmap of the 60 most incomplete proteins ordered by batch (right). Red = missing, green = observed.

7.2.2 Batch Effect Assessment and PCA Analysis

The analysis was performed before and after PCA adjustment. After adjustment, the sample distribution changed and clustering decreased. This analysis is shown in Figure 7.2.

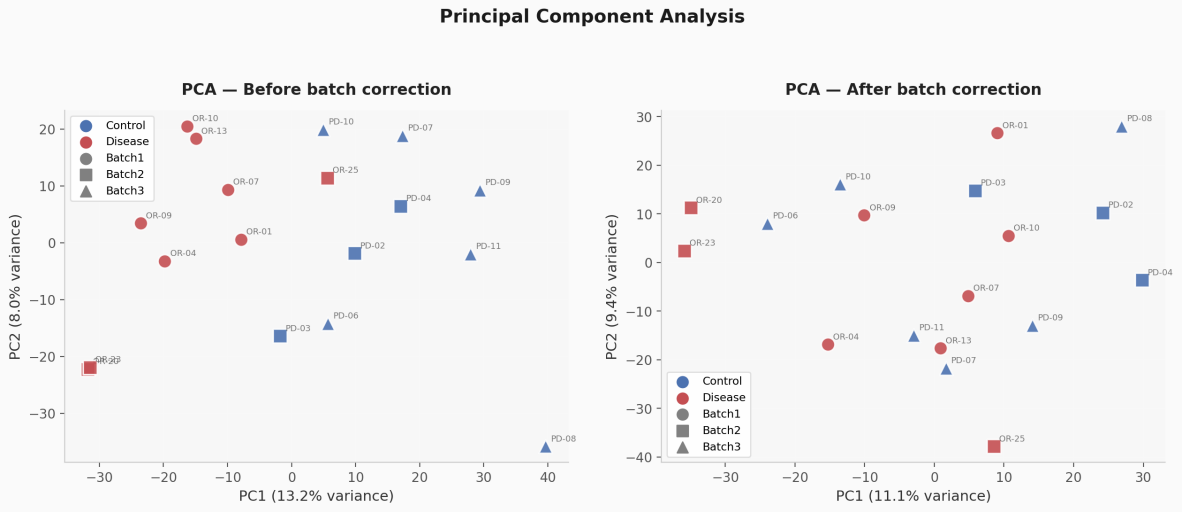


Figure 7.2.2 PCA plots before (left) and after (right) batch correction. Shapes indicate batch groups; colors indicate disease status (blue=Control, red=Disease).

7.3 Differential Analysis - Six Scenario Comparison

Differential analysis across 6 different scenarios revealed variation in the number of proteins depending on the chosen methods and correction status. The classical process (MinProb, no batch correction) identified 260 significant proteins. After batch correlation, the numbers dropped to 2-30 depending on the method. It was observed that the signals in this classical pipeline were due to technical batch variation rather than true biological differences. This variation is shown in the volcano graphs in Figure 3.

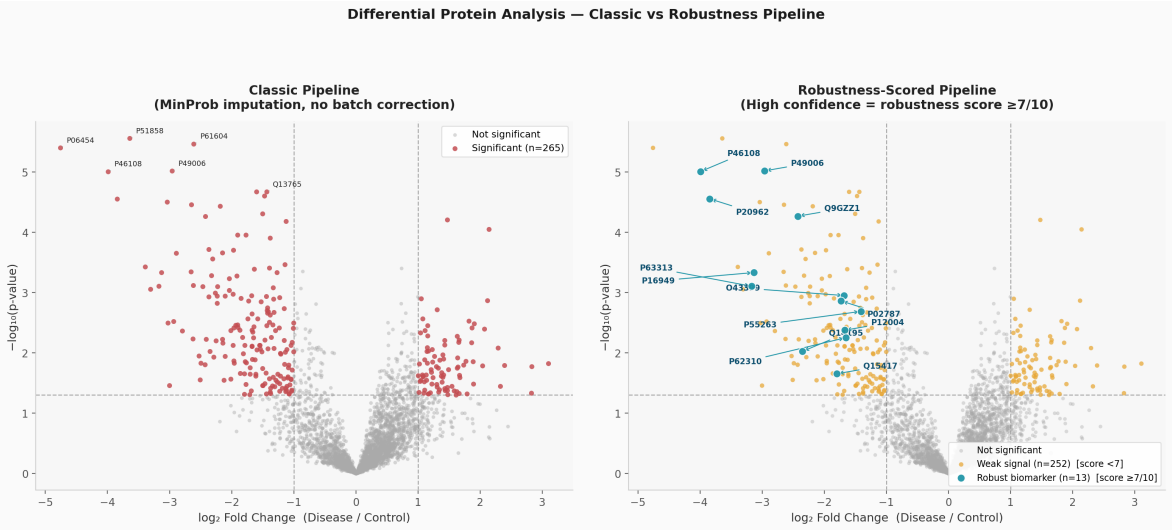


Figure 7.3 Volcano plots comparing the classic pipeline (left) and robustness-scored pipeline (right). Gray: not significant; orange: weak signal (score <7); teal: robust biomarker candidates (score >=7/10).

7.4 Robustness Score Results

Out of 3,103 analyzed proteins, 13 proteins with a high reliability score >=7/10 were classified as high reliability, 1,743 as moderate reliability, and 1,347 as low reliability or artificial. Of the 260 contaminant-filtered proteins flagged by the classical pipeline, 32 were removed as possibly artificial. The comparative table is shown in Table 2 and Figure 4.

Table 7.4 Classic pipeline vs robustness-scored pipeline comparison

	Classic Pipeline	Robustness-Scored Pipeline
Total significant proteins	260	228
High confidence (>=7/10)	not assessed	13
Moderate confidence (4-6/10)	not assessed	215
Likely artifacts removed	—	32 (12.3%)

32 proteins (12.3%) identified by the classic pipeline were reclassified as likely artifacts.

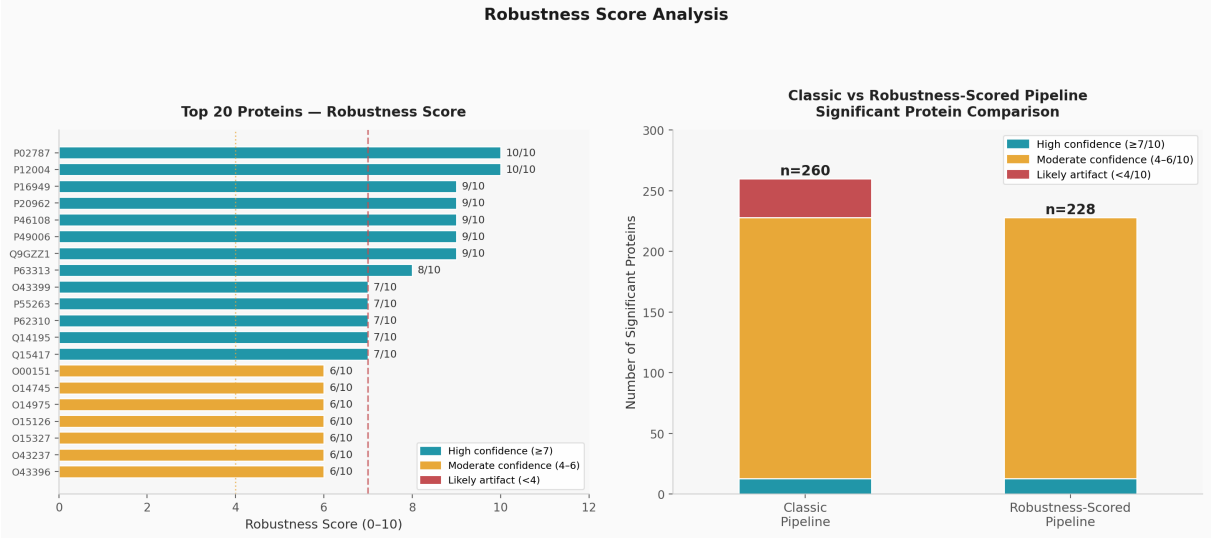


Figure 4. Left: Top 20 proteins by robustness score. Right: Comparison of classic vs robustness-scored pipeline output by confidence category.

7.5 High-Confidence Biomarker Candidates

13 proteins achieved a robustness score of 7/10 or higher and were classified as high-confidence biomarker candidates. All proteins showed negative $\log_2$FC values indicating downregulation in squamous cell carcinoma, and the candidates are listed in the table.

Table 7.5 High confidence biomarker candidates (robustness score >=7/10)

Protein	Sig. Scenarios	Missing %	Mean $\log_2$FC	Score	Category
P02787	6/6	0.0%	-1.418	10/10	High confidence
P12004	6/6	0.0%	-1.419	10/10	High confidence
P16949	4/6	22.2%	-1.916	9/10	High confidence
P20962	4/6	27.8%	-2.139	9/10	High confidence
P46108	4/6	22.2%	-2.214	9/10	High confidence
Q9GZZ1	4/6	22.2%	-1.467	9/10	High confidence
P49006	4/6	11.1%	-1.816	9/10	High confidence
P63313	4/6	33.3%	-1.615	8/10	High confidence
O43399	3/6	22.2%	-0.897	7/10	High confidence
P62310	2/6	27.8%	-0.863	7/10	High confidence
P55263	2/6	27.8%	-0.625	7/10	High confidence
Q14195	3/6	11.1%	-1.466	7/10	High confidence
Q15417	3/6	5.6%	-1.383	7/10	High confidence

7.6 Potential Artificial Effects Identified in the Classical Pipeline

The classical pipeline identified 32 proteins that were marked as significant but classified as potential artificial effects (score <4/10). These proteins showed high sensitivity to prediction methods and batch correlation. This suggests that their significance stems from analytical decisions rather than true biological differences.

In particular, two proteins (P33316 and O95858) scored 0/10, failing to meet all robustness criteria. They are listed below.

Table 7.6 Likely artifacts identified in the classic pipeline (score <4/10)

Protein	log_2FC	p-value	FDR	Robustness Score
O00193	-1.431	0.0020	0.0917	2/10
Q9Y5Z4	-1.340	0.0034	0.1207	2/10
Q86V81	-1.934	0.0035	0.1207	3/10
Q9UHD1	-1.436	0.0035	0.1207	3/10
P82979	-1.646	0.0037	0.1238	3/10
Q8WW59	-2.152	0.0059	0.1469	3/10
Q9H6Z4	-2.414	0.0060	0.1469	3/10
A8MW06;P62328	-1.674	0.0077	0.1638	3/10
P07108	-2.494	0.0148	0.2144	3/10
O15357	1.736	0.0159	0.2201	2/10
P48147	2.392	0.0160	0.2201	3/10
Q8N5J2	1.645	0.0169	0.2261	3/10
O75179;Q8IWZ3	1.213	0.0209	0.2538	3/10
Q9Y5J9	-1.616	0.0224	0.2610	3/10
P33316	-1.615	0.0240	0.2626	0/10
P42345	1.396	0.0251	0.2626	3/10
Q969P0	1.672	0.0254	0.2626	2/10
P40261	-1.391	0.0266	0.2658	2/10
Q08209;P16298;P48454	-1.165	0.0269	0.2670	3/10
P02774	-2.017	0.0272	0.2685	3/10
P26006	1.034	0.0294	0.2805	3/10
P00156	1.101	0.0298	0.2809	2/10
Q8IXM3	1.222	0.0325	0.2946	3/10
Q96IZ0	-1.770	0.0334	0.2985	2/10
P02760	-1.495	0.0348	0.3041	2/10
O94888	-1.135	0.0382	0.3177	3/10
O95858	1.345	0.0407	0.3255	0/10
P41226	1.433	0.0417	0.3255	2/10
Q16881	1.581	0.0426	0.3300	2/10
P23921	-1.379	0.0442	0.3318	2/10
Q92485	1.195	0.0456	0.3322	3/10
O00534	-1.078	0.0476	0.3345	2/10

9. Challenges and Proposed Solutions

9.1 Missing Metadata

The limited availability of metadata is one of the problems encountered in this study. Information such as how samples were prepared, the day of analysis, the status of the instrument, and clinical characteristics are not always available in detail in proteomic data.

In this study, group and batch information was obtained using the existing sample names and data structure. If RobuProt is tested with datasets containing more detailed metadata in the future, its performance can be evaluated more accurately.

9.2 Small Sample Size

In proteomic studies, the sample size is often small. The sample size in the dataset used in this study is also small. A small sample size can reduce statistical power. In this case, it may be difficult to find significance for a few proteins.

To mitigate this, we did not rely on a single analysis result. Different imputation and batch correction scenarios were performed and evaluated accordingly. Thus, proteins that were significant only in a single scenario were not highlighted. On the contrary, proteins that were consistent under different conditions were included in the biomarker protein candidate list.

9.3 Computational Cost

Using multiple imputation methods and batch correction scenarios requires more computation than classic pipeline analyses using a single scenario. This is because it analyzes 6 different scenarios for the same dataset.

Since the dataset in this study is not very large, the cost is more reasonable. However, if larger proteomic datasets are analyzed, the time will increase considerably. In the future, improving the pipeline's analysis and making it faster could alleviate this problem.

9.4 Clinical Validation Requirements

The biomarker candidates identified in this project were generated as a result of computational analysis. Therefore, it is not correct to say that these candidates can be used directly in the clinic. For clinical use, candidate proteins need to be re-validated on independent datasets and experimentally.

The purpose of RobuProt is not to make clinical diagnoses. Its purpose is to help select more reliable biomarker candidates. In the future, these candidates will need to be validated with different datasets and laboratory experiments.

9.5 Adoption Challenges

For a new bioinformatics pipeline to be usable by researchers, it is important that it is easy to understand and use. Command-line based systems may be difficult for users with little bioinformatics experience.

Therefore, it would be beneficial for RobuProt to become more user-friendly in the future. This can be achieved by providing clear documentation, sample datasets, automated reporting, and a web-based interface service.

9.6 Proposed Solutions

In summary, the proposed solutions are as follows: First, more detailed datasets can be used. Second, RobuProt should be tested with different cancer types and different proteomic datasets. Third, the code should be optimized to work with larger datasets.

Furthermore, providing a user-friendly interface for the pipeline will allow it to reach a wider audience of researchers. Finally, validating the obtained biomarker candidates with independent datasets and experimental studies increases the reliability of the method.

10. Future Applications

10.1 Multi-Omics Integration

In the future, RobuProt could be used not only with proteomic data but also to analyze genomic, transcriptomic, metabolomic, and microbiome data. Thus, biomarker candidates would not be analyzed based on a single data type. They could be selected by evaluating them together with different omics layers.

This approach could help in understanding the molecular structure of diseases from a broader perspective. Especially in complex diseases like cancer, multi-omics integration allows for stronger biological interpretations.

10.2 Machine Learning-Based Biomarker Ranking

In the future, robustness scores could be used with machine learning methods. The p-value, $\log_2$FC value, FDR value, missing value rate, and reliability in different scenarios of proteins could be used as inputs to the model. Thus, biomarker candidates would not be selected solely based on manual thresholds. They could be selected in a more advanced way thanks to machine learning. This would contribute to a more objective prioritization of candidate proteins.

10.3 Deep Learning Integration

Deep learning methods can also be added to RobuProt when working with larger proteomic datasets. Deep learning models can provide an advantage in capturing complex patterns in protein data.

However, deep learning requires a larger sample size and more stable datasets. Therefore, more advanced versions of the pipeline should be used for this stage.

10.4 Web-Based RobuProt Platform

RobuProt can serve as a web-based platform in the future. Users can easily obtain missing value analysis, batch effect evaluation, differential analysis, and robustness score results simply by uploading datasets. Automatic graph and report generation can also be added to the website. A web-based system provides convenience for users who do not want to use the command line.

10.5 Clinical Decision Support Systems

RobuProt is not yet at the level to be considered a direct clinical decision system. However, if supported by clinical validations, it will greatly assist researchers in this area in the future.

Especially in selecting more reliable biomarker candidates; It can be beneficial for models to be developed for diagnosis, prognosis, or treatment response prediction. Therefore, RobuProt can be a preliminary analysis tool for clinical decision support systems.

10.6 Precision Medicine Applications

In a personalized medicine approach, accurate interpretation of patient-specific molecular data is necessary. RobuProt-based analysis systems can enable the selection of more reliable biomarker candidates based on patient data.

In the future, such systems could be pioneers for models to be used in areas such as patient-specific risk assessment, treatment selection, and treatment response prediction.

11. Potential Impact

11.1 Scientific Impact

Statistical significance is not the only criterion in identifying biomarker candidates. Consistency between different analysis conditions has also been examined. This two-fold evaluation approach constitutes the scientific contribution of the project. This approach helps to obtain more reliable and reproducible results in proteomic studies. Furthermore, the study is important because it shows how much missing value imputation and batch correction decisions can affect biomarker results.

11.2 Clinical Impact

The most important clinical contribution is preventing the inclusion of incorrect biomarker candidates in the list. Bringing unreliable candidates to clinical trials results in a loss of time and money. RobuProt highlights more consistent candidates. Thus, it provides a more solid starting point for future clinical validation studies.

11.3 Economic Impact

Biomarker validation studies are generally costly and time-consuming processes. Since RobuProt removes candidates with low reliability at an early stage, it can reduce the need for unnecessary experimental validation studies. Therefore, it can provide economic benefits for academic laboratories, biotechnology companies, and drug development studies.

11.4 Impact on Biomarker Discovery Research

In classical biomarker discovery studies, candidate proteins are usually selected based on criteria such as p-value, FDR, or fold change. This project, however, adds a robustness score approach to these criteria and analyzes different scenarios simultaneously. This makes biomarker discovery more systematic and reliable. Furthermore, researchers can more clearly see which proteins are truly more consistent based on the score.

11.5 Impact on Computational Proteomics

This project clearly demonstrates the impact of data preprocessing decisions on results in computational proteomic analyses. Its simultaneous evaluation of different imputation and batch correction scenarios can contribute to more careful and detailed pipeline design. Therefore, RobuProt can serve as an example of reliability-focused analyses in the field of computational proteomics.

12.Discussion

This study investigated imputation and batch correction methods in LC-MS/MS proteomic analysis, which significantly influence biomarker discovery results in the literature. The single-imputation pipeline method, frequently encountered in the contaminant-free literature, yielded 260 significant proteins. However, 32 of these proteins were identified as potentially artificial in the reliability scoring method. A decrease in the number of significant proteins was observed when batch correlation and missing value methods were applied. A significant portion of the corrected analyses reflected technical batch correlation rather than true biological differences.

The two highest-scoring candidates, P02787 (Transferrin) and P12004 (PCNA), received excellent robustness scores of 10/10. Both proteins underwent six analysis scenarios in the project and were considered significant in all. P02787 (Transferrin) has previously been reported in the literature as altered iron metabolism in cancer. (PJamnongkan, W., 2019) P12004 (PCNA) is a well-known biomarker widely used in oncology. (Pandit D., 2025) The pipeline studied also yielded the same results as the biologically significant proteins in the literature. This strongly supports the validity and significance of the studied pipeline.

We see that the best and most significant candidates, as well as the two insignificant candidates, yielded results parallel to the literature. This shows that the study is consistent with the literature.

This proposed Robutness strategy creates a practical and interpretable additional layer that can be easily integrated into existing pipelines. It also establishes a reliability layer that quantifies the confidence in multiple biomarker candidates, rather than simply providing statistical data in the literature. This approach doesn't simply generate a meaningful protein list like classical pipelines; it also considers which methods and scenarios the candidate proteins remain consistent across, the percentage of missing values, whether they are preserved after batch correlation, and directional consistency. Thus, RobuProt can help identify more reliable biomarker candidates.

13. Conclusion

In this study, the RobuProt analysis pipeline, considered as a robust and layered approach, was developed to increase the reliability of biomarker discovery in LC-MS/MS-based pre-automic data and to obtain more accurate results. The study analyzed the effects of batch correlation and missing value correction on biomarker discovery. It was observed that different pre-processing strategies significantly affected the results obtained in the analysis and altered them in a way that could influence studies in the literature.The same dataset yielded different results in different analysis scenarios, producing different meaningful proteins. Therefore, it was seen that biomarker discovery based solely on statistical significance is not sufficient and reliable for biomarker selection.

Thanks to this developed RobuProt score-based approach, proteins were subjected to a score according to their reliability. Out of a total of 3103 proteins, 13 were identified as highly reliable biomarker candidates. The classical pipeline, on the other hand, presented the number of meaningful proteins as 32.

Final of the project, the developed RobuProt pipeline contributed to reducing false positive biomarker discoveries caused by missing values and batch effects, and created an additional layer of reliability in biomarker selection. Furthermore, since it could influence future studies, it is anticipated that it will be further developed by applying it to different proteomic datasets.

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